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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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ASSESSMENT -2 – TECHNICAL PRESENTATION

CSA0302- DATA STRUCTURES

TOPIC 1 - An operating system manages multiple processes waiting for CPU allocation in a time-sharing environment. Recommend an appropriate data structure and justify based on time and space efficiency.

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Problem Identification

- **Problem Statement**

- In a time-sharing operating system, multiple processes wait for CPU execution at the same time.
- The operating system must:
 - Allocate CPU fairly to each process.
 - Prevent starvation.
 - Execute processes in an organized manner.
 - Efficiently manage waiting processes.
- The challenge is selecting a suitable data structure that provides efficient process scheduling with minimal time and memory usage.

Recommended Data Structure

- **Queue (Circular Queue)**
- A Queue is a linear data structure that follows the **FIFO (First In, First Out)** principle.
- In CPU scheduling:
- New processes are inserted at the rear.
- The CPU executes the process at the front.
- If a process is not completed within its time slice, it is moved to the rear of the queue.
- This mechanism is commonly used in **Round Robin Scheduling**.

Working of Queue

- **Process Execution Flow**

- A process enters the Ready Queue.
- CPU selects the first process.
- The process executes for one time quantum.
- If completed, it leaves the queue.
- Otherwise, it is moved to the rear.
- The next process receives CPU time.
- This cycle continues until all processes are completed.

Why Queue is Suitable

- **Justification**

- Queue is the most suitable data structure because:
- Processes are executed in arrival order.
- Provides fair CPU allocation.
- Supports Round Robin Scheduling.
- Easy insertion and deletion operations.
- Reduces waiting time.
- Efficient memory utilization

Time and Space Efficiency

Operation	Complexity
Insert Process	$O(1)$
Remove Process	$O(1)$
Access Front Process	$O(1)$
Space Complexity	$O(n)$

Advantages

- **Advantages of Queue**
- Simple implementation.
- Fast enqueue and dequeue operations.
- Fair scheduling.
- Supports multitasking.
- Prevents starvation.
- Widely used in modern operating systems.

Applications

- **Real-World Applications**
- CPU Scheduling
- Printer Queue
- Keyboard Buffer
- Process Scheduling
- Network Packet Scheduling
- Task Scheduling

Conclusion

- Queue is the most appropriate data structure for CPU scheduling in a time-sharing operating system.
- Its FIFO nature ensures fair execution of processes while maintaining $O(1)$ insertion and deletion time, making it efficient in both performance and memory usage.